Table 1. Per-step timing at 160M, 780M, and 820M scale (4× H100 GPU, averaged over 180 steps after warm-up). SPECTRA overhead shrinks from ∼17–19% at 160M to ∼2% at 780M/820M as the forward/backward pass dominates at larger scale. D-Muon has comparable overhead to SPECTRA due to its NS-based orthogonalization (∼9% at 160M, ∼1–2% at 780M/820M). SOAP is the most expensive optimizer, adding +49% at 160M and +21% at 780M vs. AdamW, due to its larger memory footprint (requiring more accumulation steps), periodic eigendecomposition, and per-step gradient projection.

	160M			780M			820M		
Method	Total (ms)	Opt Step (%)	Overhead	Total (ms)	Opt Step (%)	Overhead	Total (ms)	Opt Step (%)	Overhead
AdamW	200.7	1.0	—	6550	0.2	—	4412	0.2	—
+ Pre-Clip	232.6	13.7	+15.9%	6640	1.4	+1.4%	4480	1.7	+1.5%
+ SPECTRA	238.3	15.6	+18.7%	6654	1.7	+1.6%	4504	2.4	+2.1%
Signum	203.2	1.9	—	6536	0.3	—	4408	0.5	—
+ SPECTRA	241.2	17.1	+18.7%	6667	1.9	+2.0%	4514	2.6	+2.4%
AdEMAMix	207.9	4.3	—	6575	0.7	—	4456	1.3	—
+ SPECTRA	242.5	18.2	+16.6%	6691	2.2	+1.8%	4568	3.3	+2.5%
MARS	210.8	5.5	—	6591	0.8	—	4464	1.4	—
+ SPECTRA	248.3	18.1	+17.8%	6699	2.4	+1.6%	4553	3.4	+2.0%
D-Muon	220.5	8.8	—	6637	1.4	—	4496	2.1	—
SOAP	299.1	33.2	—	7917	7.9	—	4868	11.0	—

Opt Step = optimizer step fraction of total step time. Overhead = overhead vs. corresponding base optimizer.

Table 2. Fixed-time budget comparison at 780M scale. Baseline optimizers are given 5–20% more training steps (16800–19200 vs. 16000) to compare with SPECTRA's 2% per-step overhead. SPECTRA variants trained for 16000 steps are shown in the last column.

	Baseline (extended steps)				SPECTRA
Optimizer	16000	+5% (16800)	+10% (17600)	+20% (19200)	16000
AdamW	2.608	2.605	2.600	2.590	2.582
Signum	2.600	2.594	2.589	2.580	2.573
AdEMAMix	2.591	2.585	2.580	2.571	2.574
MARS	2.598	2.592	2.587	2.579	TBD
D-Muon	2.595	2.589	2.584	2.574	—

Table 3. SOAP vs. SPECTRA-Signum at larger scales (final validation loss). SOAP is tuned over three learning rates. Despite SOAP requiring ∼19% more time per step at 780M (+8% at 820M) and nearly 2× more memory than SPECTRA-Signum, it achieves comparable results at 780M and worse results at 820M.

Method (lr)	780M	820M
SOAP (10^{-3})	2.586	2.606
SOAP (3×10^{-3})	**2.571**	2.602
SOAP (5×10^{-3})	2.574	2.603
SPECTRA-Signum (10^{-3})	2.573	**2.594**

Table 4. Sensitivity to clipping threshold c (160M, WSD schedule, lr=10^{-3}). Baseline AdamW without SPECTRA: 2.958. Performance is flat from $c = 5$ to $c = 50$ (all within 0.004 of the optimum). Only $c = 1$ degrades meaningfully due to over-aggressive clipping.

c	1	5	10	20	50
Val Loss	2.975	2.940	**2.936**	**2.936**	2.940

Table 5. Number of NS iterations (160M, WSD schedule, lr=10^{-3}). (bfloat16 has only 7 bits of mantissa. Additional iterations beyond convergence don't improve accuracy–they accumulate round-off errors. Therefore, we use float32 for NS=15 instead.)

NS Iters	5 (bf16)	10 (bf16)	15 (float32)
Val Loss	2.939	2.936	2.937

Table 6. SPECTRA applied to Mars across three scales (final validation loss) with $c = 10$ shows consistent improvement over Mars.

	160M	780M	820M
MARS	2.984	2.597	2.614
SPECTRA-MARS	**2.962**	TBD	**2.589**

Table 7. The evaluation results (accuracy $\times 100\%$) of the 820M model pretrained on Fineweb-Edu with 16.8B tokens (5-shot with lm-evaluation-harness).

Optimizer	**ARC-c**	**ARC-e**	**BoolQ**	**PIQA**	**HellaSwag**	**WinoGrande**	**OBQA**	**λ-OpenAI**	**Sciq**	**MMLU**	**Average**
Mars	34.98	67.80	57.25	70.08	46.87	52.17	34.20	**29.38**	89.60	25.37	50.77
D-Muon	36.60	67.47	53.73	69.31	46.69	52.72	34.60	28.84	**90.90**	25.08	50.59
Signum	34.56	65.66	56.02	69.64	45.01	52.25	34.20	28.04	87.30	25.75	49.84
Spectra-Signum	35.58	**68.81**	**58.10**	**70.46**	47.04	**54.46**	35.60	29.36	90.30	**26.44**	**51.62**
AdamW	33.45	66.84	56.09	69.21	45.65	50.99	35.20	25.54	90.80	23.86	49.76
Spectra-AdamW	35.67	66.96	55.41	68.82	46.57	52.09	**36.40**	26.74	89.20	25.14	50.30
AdEMAMix	35.41	65.95	43.43	70.08	46.43	53.20	34.40	28.51	88.30	24.84	49.05
Spectra-AdEMAMix	**36.86**	68.81	53.94	69.48	**47.09**	53.43	36.40	28.26	89.40	26.23	50.99

Table 8. Short training (8B tokens or 8k steps) of 1.5B muP models.

Optimizer	**LR**	**Final Val Loss**
SOAP	5×10^{-3}	2.653
D-Muon	2×10^{-3}	2.658
MARS	2×10^{-3}	2.643
AdEMAMix	2×10^{-3}	2.647
	3×10^{-3}	2.642
Spectra-Signum	1×10^{-3}	2.647
	2×10^{-3}	2.642
	3×10^{-3}	2.638
	5×10^{-3}	2.643
Spectra-AdEMAMix	1×10^{-3}	2.654
	2×10^{-3}	2.640
	3×10^{-3}	**2.635**

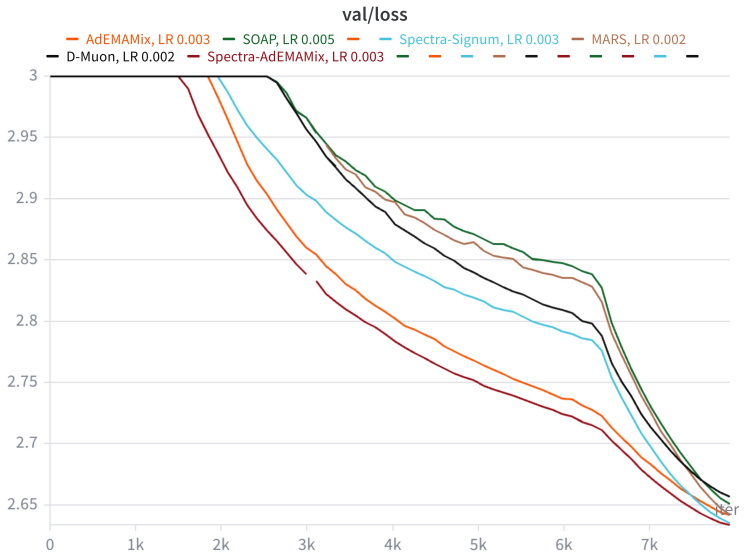

Figure 1. Among all optimizers, our SPECTRA versions are shown to be superior. Throughout the entire training run, we see how Spectra-AdEMAMix improves upon the original AdEMAMix. See Table 8 for final validation loss values.

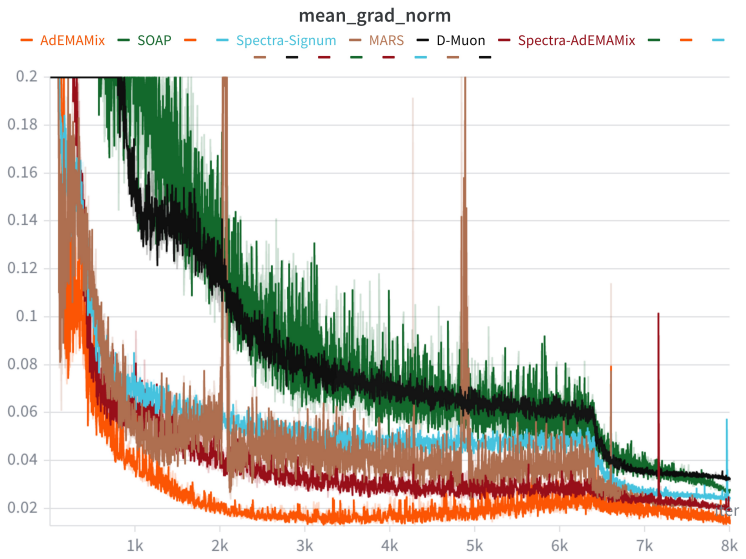

Figure 2. Gradient norm patters for all optimizers at 1.5B scale with muP.

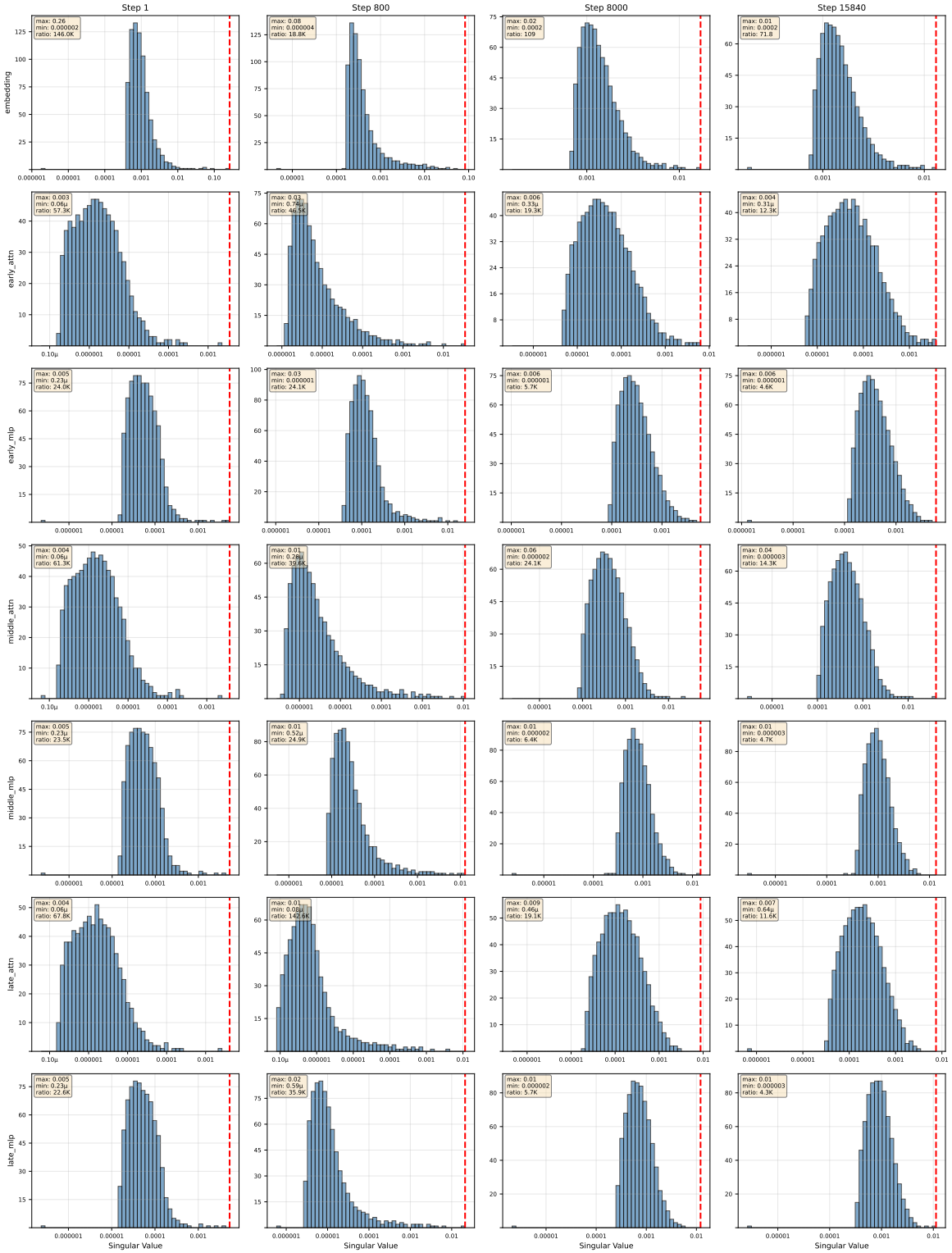

Figure 3. Singular value distribution of raw stochastic gradients for the 124M-parameter GPT with muP trained with AdamW. The heavy-tailed 'spectral outlier' phenomenon is less significant than 124M-llama model without muP.

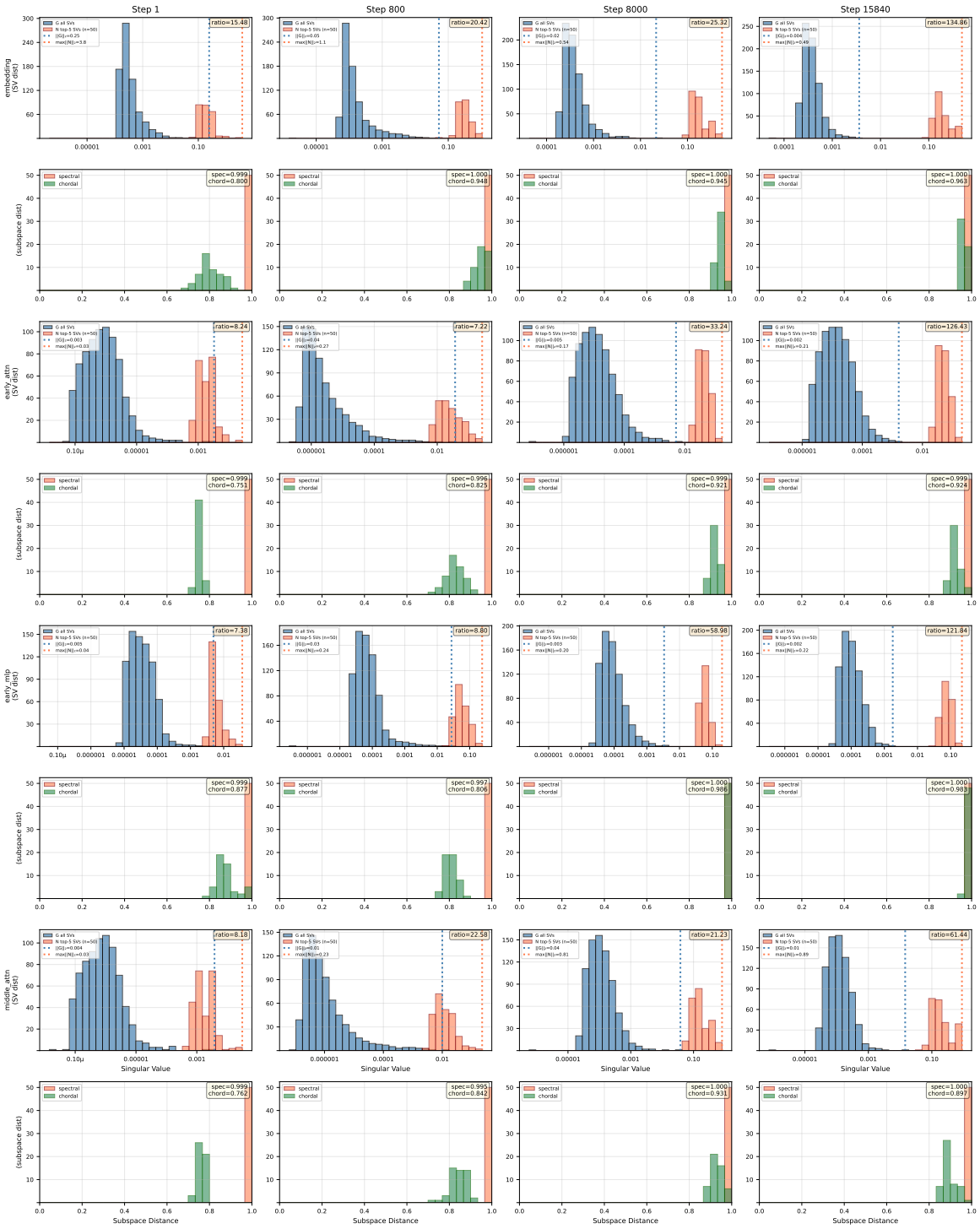

Figure 4. For each layer, we report the singular value distribution of large-batch gradient $\mathbf{G}$ (blue bar) and top-5 singular value distribution of the noise matrices $\{\mathbf{N}_i\}_{i=1}^{50}$ (red) in the first row. In the second row, we report the distribution of both the spectral and chordal subspace distances of the top-5 principal directions between the noise matrices and the large-batch signal. Results are shown for representative layers of the 124M GPT model with muP (AdamW) at four training stages 0%, 5%, 50%, and 99%. The presence of "sparse spectral spikes" in the noise appears mostly only in the end of the training.